Meng'en Qin

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Summary

Hi, I'm Meng'en Qin, a statistics graduate from Henan University. My research journey began at the Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, where I explored interpretable visual representation learning and its interdisciplinary applications. Since then, I've worked on interpreting the internal mechanisms of MLLMs/LLMs, particularly their reasoning and hallucination behavior. I'm currently working with Prof. Ruize Han and Distinguished Prof. Song Wang at Shenzhen University of Advanced Technology.

Education

School of Mathematics and Statistics, Henan University — *Kaifeng, China*

B.S. in Statistics — *Sep 2021 - Jun 2025*

- Faculty Ranking: top 8% and GPA: 87/100

Research Experience

Shenzhen University of Advanced Technology — *Shenzhen, China*

Research Assistant — *Sep 2025 - Sep 2026*

- reasoning and hallucinations of MLLMs/LLMs (working with Prof. Ruize Han, Prof. Youlu Xing and Distinguished Prof. Song Wang)

Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms — *Henan, China*

Research Intern — *Nov 2023 - Jan 2025*

- interpretable visual representation learning and its interdisciplinary applications (work with Prof. Xiaohui Yang)

Honors & Awards

Nov 2023	**Provincial First Prize**, China Undergraduate Mathematical Contest in Modeling	*Henan, China*
Aug 2023	**National Second Prize (Top 0.5%)**, National College Student Statistical Modeling Competition	*China*
Aug 2023	**National Third Prize**, National College Students Social Practice and Science Contest on Energy Saving & Emission Reduction	*China*

Papers

MLLMs Hallucinate when Information Distribution Drifts in Synergy Heads — *arXiv:2609.09206*

Meng'en Qin, Junye Chen, Jucheng Liu, Yinchen Liu, Youlu Xing, Song Wang, Ruize Han† — *2026*

Adaptive Compressed Residual: Redundancy-Aware Mechanism for Compact Multimedia Representation Learning — *revision resubmitted to TMM*

Meng'en Qin*, Yu Song*, Jing Fang, Deyu Meng, Xiaohui Yang†, Lili Yang† — *2026*

Training-Adaptive Convolutional Sparse Coding via Information Bottleneck for Robust Visual Representation — *arXiv:2609.19122*

Meng'en Qin, Yinchen Liu, Mingxuan Cui, Youlu Xing† — *2026*

A^3-FPN: Asymptotic Content-Aware Pyramid Attention Network for Dense Visual Prediction — *Pattern Recognition*

Meng'en Qin, Yu Song, Quanling Zhao, Xiaodong Yang, Yingtao Che, Xiaohui Yang† — *2026*

StomaD^2: An All-in-One System for Intelligent Stomatal Phenotype Analysis via Diffusion-Based Restoration Detection Network — *arXiv:2604.18632*

Quanling Zhao*, Meng'en Qin*, Yanfeng Sun, Yuan Miao†, Xiaohui Yang† — *2025*

Significance and Stability Analysis of Genotype-Environment Interaction using GxEStat — *submitted to Journal of Applied Statistics*

Meng'en Qin, Zhe Li, Hui Huang, Xihong Liu† — *2026*

OS-MSWGBM: Intelligent Analysis of Organic Synthesis Based on Multiscale Subtraction Weighted Network and LightGBM *MATCH Commun. Math. Comput. Chem.*

Lanfeng Wang*, Yanhui Guo*, Zelin Zhang, Meng'en Qin, Zixin Li, Xiaoli Sun, Xiaohui Yang† *2025*

OCS-TGBM: Intelligent Analysis of Organic Chemical Synthesis Based on Topological Data Analysis and LightGBM *MATCH Commun. Math. Comput. Chem.*

Yanhui Guo*, Lichao Peng*, Zixin Li, Meng'en Qin, Xue Jiao, Yun Chai and Xiaohui Yang† *2024*

Patents

Information Intelligent Compensation and Compression Based on Progressive Fusion of Spatial and Channel Features *CN119131792B*

Lichao Peng, Meng'en Qin, Xiaohui Yang, Chen Miao, Yanfeng Sun, Han Qiao *2025*

Coal and Gangue Intelligent Recognition Based on Selective State Space Equation *CN119048740A*

Xiaohui Yang, Meng'en Qin, Yanni Zhang, Zheng Ge, Yunfei Bai *2024*

Intelligent Analysis Method of Stomatal Phenotype of Crop Non-destructive Leaves Based on HDIoU *CN118691972A*

Lichao Peng, Meng'en Qin, Xiaohui Yang, Chen Miao, Yanfeng Sun *2024*

Multimodal Data Integration Based on Convolutional Sparse Coding and Optimal Transport *CN117763499B*

Xiaohui Yang, Jingjing Li, Yuan Feng, Lichao Peng, Yu Song, Meng'en Qin *2023*

Software

GxEStat *R*

Meng'en Qin, Xiaohui Yang, Yanfeng Sun, Chen Miao *2025*

- GxEStat is a user-friendly R GUI package designed for statistical analysis of genotype-by-environment interaction. Additionally, GxEStat is packaged as a standalone desktop application using Electron, and you can download the Windows release here.

Vehicle Trajectory Analysis Based on Path Planning *Matlab*

Meng'en Qin *2024*

- This software integrates functionalities such as speed calculation, road travel time analysis, traffic flow estimation, congestion visualization, and so on, which are suitable for urban traffic management, intelligent transportation systems, and autonomous driving path research.

Extracurricular Activities

Stellar Corps, Young Sprout Association *Yuqing Middle School, Guizhou, China*

Vice Captain and Teaching Volunteer *Jan 2023 - Dec 2024*

- I joined the Young Sprout Association of Henan University and served as vice captain of Stellar Corps (chapter for high school students). Our team brought together members from diverse disciplinary backgrounds, including medicine, chemistry, physics, education, business, and statistics. We collaborated with Shanghai Lingqing Philanthropy Development Center and co-launched a long-term volunteer project with Yuqing Middle School in Baini Town, Yuqing County, Guizhou Province. We spent most vacations (winter, summer, and national vacations) on volunteer teaching activities, student mentoring, and course design. Our team has created a series of engaging extracurricular programs that integrated academic concepts with real-world challenges.

Debate Team of School of Mathematics and Statistics, Henan University *Kaifeng, China*

Fourth Debater *Oct 2021 - Dec 2022*

- I served as the fourth debater on the debate team of School of Mathematics and Statistics. As the closing speaker, I was responsible for delivering final rebuttals, synthesizing arguments, and constructing impactful conclusions.